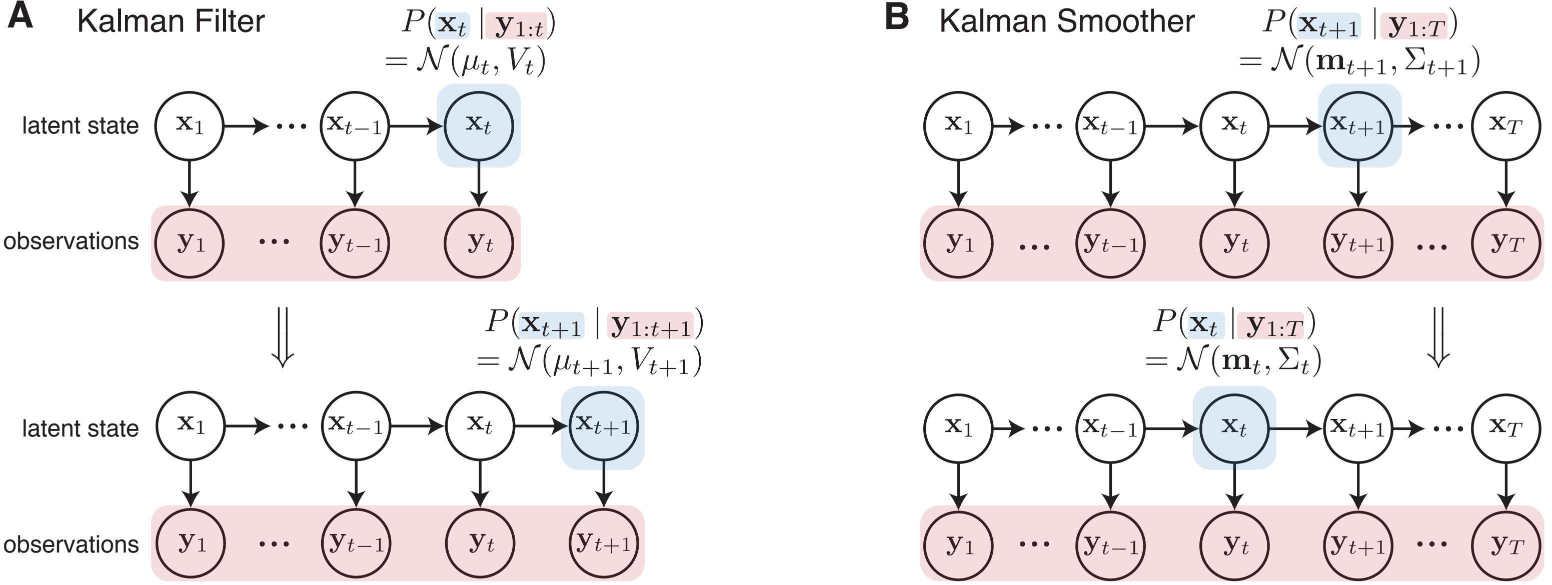

A
Kalman Filter
$P(\mathbf{x}_t \mid \mathbf{y}_{1:t}) = \mathcal{N}(\mu_t, V_t)$
latent state
observations
$\mathbf{x}_1$ $\mathbf{x}_{t-1}$ $\mathbf{x}_t$ $\mathbf{x}_{t+1}$
$\mathbf{y}_1$ $\mathbf{y}_{t-1}$ $\mathbf{y}_t$ $\mathbf{y}_{t+1}$
$P(\mathbf{x}_{t+1} \mid \mathbf{y}_{1:t+1}) = \mathcal{N}(\mu_{t+1}, V_{t+1})$
B
Kalman Smoother
$P(\mathbf{x}_{t+1} \mid \mathbf{y}_{1:T}) = \mathcal{N}(\mathbf{m}_{t+1}, \Sigma_{t+1})$
$\mathbf{x}_T$ $\mathbf{y}_T$
$P(\mathbf{x}_t \mid \mathbf{y}_{1:T}) = \mathcal{N}(\mathbf{m}_t, \Sigma_t)$